Sakina Fatima, PhD. | sakina725@gmail.com | +1(343)463-1255 | Toronto, Canada

Education

Jan 2021 – Dec 2024
University of Ottawa – Canada
PhD in Computer Science

Sept 2018 – Aug 2020
Maynooth University – Ireland, University of St Andrews – United Kingdom
Erasmus Mundus Joint MSc in Dependable Software Systems

Feb 2013 – Jan 2017
Comsats University – Pakistan
BSc in Computer Science

Work Experience

May 2025 – Present
Fujitsu Research – Canada
AI Researcher
- Accelerating LLM inference through model level optimizations, including pruning and scalable GPU execution
- Exploring reinforcement learning–based compiler optimization and embedding techniques within the MLIR ecosystem

Jan 2025 – April 2025
Nanda Lab, University of Ottawa – Canada
Research Fellow
- Fault Localization, tracking execution traces with Control Flow Graphs (CFGs), and automatic repair for system level test code on domain-specific industrial dataset using Deep Learning models i.e. LLMS (prompt engineering, fine-tuning and RAG with LangChain- in collabration with Huawei Technologies Co., Ltd.)

Jan 2021 – Dec 2024
Research Assistant
PhD. Advisor: Dr Lionel Briand
- Analysing **Flaky Tests** in Regression Testing and Continuous Integration
- Developing **Black-Box** solutions and tools to detect, categorize and repair Flaky Tests using traditional machine learning models and **Pre-trained Large Language Models**
- Generating a larger, open-source dataset of Flaky Tests

July 2023 – Oct 2023
Microsoft Research Lab (RiSE Group), Microsoft Corporation – Redmond, USA
PhD Research Intern
- Developing solutions for Software Code Generation and Repair using **Large Language Models** (LLMs) including GPT, CodeT5, CodeGen
- Using **Reinforcement Learning** (RLHF, PPO, DPO) to further enhance the Performance of LLMs

Jan 2021 – Apr 2021
Dept of Engineering, University of Ottawa – Canada
Lead Teaching Assistant
- Course: *Introduction to Programming II*
- Developed and presented tutorials and educational videos; held office hours; evaluated reports, conducted examination
- Developed **JUnit Test Cases** for automatic grading of submitted assignments

Jan 2020 – June 2020
SACHI Research Group, University of St Andrews – United Kingdom
Research Assistant
- Analysed video images from drones to locate dolphin schools using different techniques of **image analysis** and **ML (A project of Inter-American Tropical Tuna Commission (IATTC))**

July 2019 – Aug 2019
Dept of Computer Science, Maynooth University – Ireland
Software Engineer Intern
- Worked on developing 'Collaborative Annotation and Pedagogy Tool' tool for university using **Node.Js**, **Express**, **Ajax** and **MongoDb**

Aug 2017 – Sept 2018

International Business Machines Corporation (IBM) – Pakistan
Software Consultant

- Performed Quality Assurance of **ETL** processes with **IBM RTC** and **RQM Tool**
- Worked with **SQL** queries and Business Intelligence team for Quality Assurance of IBM Cognos Reports
- Arranged around 30 **IBM Cloud** workshops and Hackathons in universities and Incubation centres, all over Pakistan to promote IBM Cloud and technologies

Publications

- P. Okanovic, S. Deshmukh, G. Kwasniewski, Y. Zhu, H. Fujii, **S. Fatima**, M. Besta, K. Katayama, T. Honda, Y. Nagasaka, T. Hoefler, **BLaST: High Performance Inference and Pretraining using BLock Sparse Transformers**, Submitted, IEEE International Parallel & Distributed Processing Symposium, IPDPS 2026 (Under Review)
- A. S. Yaraghi, G. Gharachorlu, **S. Fatima**, L. C. Briand, R. Wan, R. Gao, **Black-Box Test Code Fault Localization Driven by Large Language Models and Execution Estimation**, Submitted, IEEE Transactions on Software Engineering (Under Review)
- **S. Fatima**, H. Hemmati, L. Briand, **FlakyFix: Using Large Language Models for Predicting Flaky Test Fix Categories and Test Code Repair**, (Accepted, IEEE Transactions on Software Engineering, TSE, Accepted as Journal First at **ICST, 2025**)
- S. Chakraborty, G. Ebner, S. Bhat, S. Fakhoury, **S.Fatima**, S. Lahiri, N. Swamy, **Towards Neural Synthesis for SMT-Assisted Proof-Oriented Programming**, (Accepted, International Conference on Software Engineering, **ICSE, 2025**, Won **ACM SIGSOFT Distinguished Paper Award** Work done as part of Microsoft Research Internship)
- **S. Fatima**, T.A. Ghaleb, L. Briand, **Flakify: A Black-Box, Language Model-based Predictor for Flaky Tests**, (Published, IEEE Transactions on Software Engineering, TSE, Presented as Journal First at **ICSE, 2023**)
- L. McMillan, **S. Fatima**, C.S. Oedekoven, C. Faustino, C.E. Lennert-Cody, S.T. Buckland, D. Harris-Birtill, **Triton a mixed machine-learning strategy for classifying cetaceans in low-resolution video**, (Submitted, IEEE Transactions on Pattern Analysis and Machine Intelligence)
- C.S. Oedekoven, L. McMillan, **S. Fatima**, D. Harris-Birtill, C. Faustino, L. Marshall, S.T. Buckland, and C.E. Lennert-Cody, **Trial survey for eastern tropical Pacific dolphins**: Project report, IATTC Special Report, No. 24

Honours and Awards

- **Mitacs Canada Grant of $35000** yearly, 2021-2025
- **International Admission Scholarship and International Doctoral Scholarship of $18000** yearly by University of Ottawa, 2021-2025
- **French Government Medal, and NUI Prize of €1000 for Distinction in Collaborative Degrees**, 2019
- **Best Thesis Prize** in Summer School at University of Lorraine, Nancy France, 2019
- **Erasmus Mundus Scholarship of €49000** by European Commission for Joint Master's Degree 2018-2020

Leadership and Extracurricular

Sept 2022 – May 2023

Google Developer Group (GDG)-Ottawa Chapter
Team Lead
https://gdg.community.dev/gdg-ottawa/

July 2021 – June 2023

Erasmus Mundus Student and Alumni Association (EMA)
Director and Board Member
https://www.em-a.eu/mb-21-23

Apr 2022 – April 2023

Faculty Council of University Of Ottawa
Member

Jan 2016 – Dec 2016

Network of Organizations Working with People with Disabilities (NOWPDP), Pakistan
Student Ambassador

DISTINGUISHED PAPER AWARD

This certificate is proudly presented to

Saikat Chakraborty, Gabriel Ebner, Siddharth Bhat,
Sarah Fakhoury, Sakina Fatima, Shuvendu K. Lahiri, Nikhil Swamy

For the paper "Towards Neural Synthesis for SMT-assisted Proof-Oriented Programming"

Awarded at the IEEE/ACM 47th International Conference on Software Engineering (ICSE 2025)

Research/Main Track

Sun 27 April – Sat 3 May 2025 – Ottawa, Ontario, Canada

Lionel Briand	***Timothy Lethbridge***		***David Lo***	***Corina S. Pasareanu***
General Co-Chair	*General Co-Chair*		*Program Co-Chair*	*Program Co-Chair*

Ms Sakina Fatima
Best First Year Student on Erasmus Mundus Double MSc in Dependable Software Systems
Department of Computer Science

10 January 2020

REF: PRIZES AND SCHOLARSHIPS AWARDS CEREMONY – 12th February 2020

Dear Sakina,

Every year Maynooth University holds a Prizes and Scholarships Awards Ceremony to present students with prizes for academic achievement. You are among the prize recipients for the academic year 2018/2019. We wish to congratulate you on this achievement.

The Prizes and Scholarships Awards Ceremony will be held over two evenings, Tuesday 11th and Wednesday 12th February 2020. Your ceremony will be held on **Wednesday evening 12th at 7pm in the Aula Maxima, South Campus, Maynooth University**. I enclose herewith three invitation tickets to the ceremony, one for you and two guests. All prize recipients should register with a member of staff in the foyer of the Aula Maxima before 6.00pm at the latest and you will then be shown to your seat in the prize recipients' section of the hall.

Don't forget your invitation tickets as admission will be by ticket only. Please ensure that you check the exact date of your particular ceremony on your tickets and our website https://www.maynoothuniversity.ie/conferring/prizes-and-awards. You and your guests should be seated in the Aula Maxima by 6:15pm to allow for a prompt start. The ceremony will be followed by a reception in Pugin Hall.

If you will not be in a position to attend, I would be grateful if you could let me know by email (Graduation@mu.ie), or by telephone (01-474-7782), before Friday 7th February.

Congratulations once again.

Yours sincerely,

Brenda Wilson
Maynooth University Conferring Office
01-708-6583
Graduation@mu.ie

Ollscoil na hÉireann

National University of Ireland

This is to certify that on the 26th of November 2019

Sakina Fatima

Maynooth University

was awarded the

French Government Medal and NUI Prize

for Distinction in Collaborative Degrees

Maurice Manning

Chancellor

November 2019

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Erasmus+

2019 Erasmus Mundus DEPEND Programme award for

Best Project

Maynooth University

Sakina Fatima

Winner

Dr Joseph Timoney
DESEM Programme Co-Ordinator, Maynooth University

UNIVERSITÉ DE LORRAINE

Maynooth University
National University of Ireland Maynooth

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Erasmus Mundus Association

Certificate
OF PARTICIPATION

This certificate is awarded to

Sakina Fatima

for participation in the 12th General Assembly of Erasmus Mundus Association
held on 23-25 February, 2019 at Universität Wien, Austria.

Pavan Kumar Sriram
Erasmus Mundus Association, President

Lucia Salome Loposova
Erasmus Mundus Association, Vice-President

THINK

BEFORE YOU PRINT

Erasmus
Mundus
Association

Maynooth University
National University of Ireland Maynooth

Roinn Ríomheolaíochta Ollscoil Mhá Nuad
Maynooth University Department of Computer Science

1 June 2018

TO WHOM IT MAY CONCERN

Ms Sakina Fatima has been accepted as a student on the Erasmus Mundus Joint Masters Course (EMJMD) MSc in Advanced Systems Dependability (DEPEND) on a 'Partner Country Scholarship'. The funds that this student will receive for this two year programme total €49,000 and are detailed below. The payment of the funds will be administered by Maynooth University as part of the DEPEND Consortium.

The monthly allowances of the scholarship for living costs are paid to the Student in twenty four monthly instalments of €1000 each. The payments of the monthly instalments should begin within 30 days of the commencement of the first academic year 24th September 2018, but not before the completion of registration at the first host University. These monthly instalments are paid by bank transfer to the Student's personal bank account details of which must be provided at time of registration. The contribution to the Student's travel will be €3,000 per annum and a once off installation payment of €1,000 to assist with any other type of personal costs associated with the students relocation, will also be given. These payments should be paid upon the arrival and registration of the Student at their host University at the beginning of each academic year where applicable and following signature of the Student Agreement.

The DEPEND Consortium retains the contribution of €4,500 per semester of the scholarship to cover the 'participation costs' to the DEPEND Erasmus Mundus Joint Masters programme. These participation costs include any compulsory administrative and operational costs related to the participation of the Student in the DEPEND Erasmus Mundus Joint Masters programme (e.g. admission, registration, university tuition, course exams, thesis examination, library and computer access, insurance coverage, etc.). The insurance scheme selected by the Coordinating Institution covers adequately the Student in case of accident, injury, hospitalisation and illness while participating in the DEPEND Erasmus Mundus Joint Masters Programme. Any other costs that may be charged in addition (e.g. for participation in fieldwork activities and summer sessions), being compulsory or voluntary, are communicated to the Student at the application stage. These costs and any other costs, such as document application, visa and residence permits, books, local transportation, travel to the host universities and accommodation, including the travel and accommodation expenditures to participate in the two summer schools are **not** covered by these 'participation costs'.

If you have any further queries please do not hesitate to contact me at mscdepend@mu.ie.

Prof Adam Winstanley
DEPEND Consortium Office

Roinn Ríomheolaíochta Ollscoil Mhá Nuad, Ollscoil Mhá Nuad, Maigh Nuad, Co. Chill Dara, Éire.
Maynooth University Department of Computer Science, Maynooth University, Maynooth, Co. Kildare, Ireland.

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